Section B: Practical Tasks

1. Define UserProfile Model: Create a class in models.py inheriting from models.Model. Define fields such as CharField for the username, IntegerField for age, and BooleanField for the is_public status.

Ans=

```python
from django.db import models

class UserProfile(models.Model):
    username = models.CharField(max_length=100)
    age = models.IntegerField()
    is_public = models.BooleanField(default=True)

    def __str__(self):
        return self.username
```

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USERNAME	AGE	IS PUBLIC
alice	25	✔

2. create ModelForm & Validation: Implement a UserProfileForm extending forms.ModelForm. Add a clean_age() method to enforce custom constraints, such as ensuring the user is over 13 years old before allowing profile creation.

Ans=

models.py

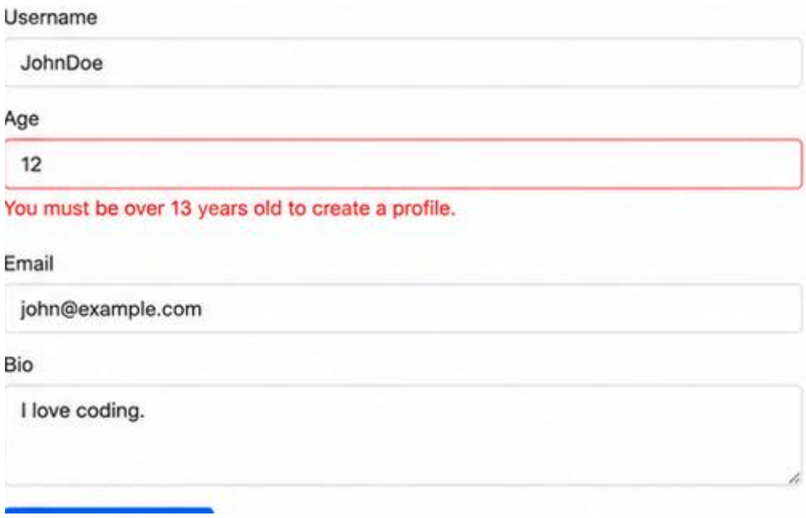
```python
from django.db import medels

class UserProfile(models.Model):
    username = models.CharField(max_length=100)
    age = models.IntegerField()
    email = models.EmailField()
    bio = models.TextField(blank=True)

    def __str__(self):
        return self.username
```

forms.py

```python
from django import forms
from .models import UserProfile

class UserProfileForm(forms.ModelForm):
    class Meta:
        model = UserProfile
        fields = ['username', 'age', 'email', 'bio']

    def clean_age(self):
        age = self.cleaned_data.get('age')
        if age is not None and age <= 13:
            raise forms.ValidationError(
                "You must be over 13 years old to create a profile."
            )
        return age
```

views.py

```python
from django.shortcuts import render, redirect
from .forms import UserProfileForm

def create_profile(request):
    if request.method == 'POST':
        form = UserProfileForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('success')
        else:
    form = UserProfileForm()
    return render(request, 'create_profile.html', {'form': form})
```

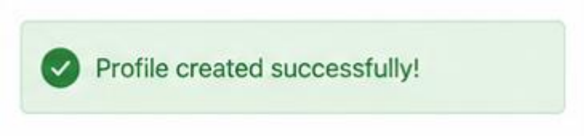


3. Django Views for Persistence: Develop a function-based or class-based view to handle POST requests. Use form.is_valid() to check inputs and form.save() to commit the new profile data directly to the database.

Ans=

```
from django.shortcuts import render, redirect
from .forms import UserProfileForm

def create_profile(request):
    if request.method == 'POST':
        form = UserProfileForm(request.POST)
        if form.is_valid():
            form.save()  # saves to DB
            return redirect('profile_success')
        else:

    form = UserProfileForm()  # for GET request

  return render(request, 'create_profile.html', {'form': form})
```

```
from django.urls import path
from . import views

urlpattmers = [
    path('profile/create/', views.create_profile, name='create_profile'),
    path('profile/success/', views.profile_success, name='profile_success'),
]
```



4. Render Profiles with DTL: Retrieve all profiles using UserProfile.objects.all() and pass them to a template. Use Django Template Language (DTL) loops ({% for profile in profiles %}) to display the username and age in an HTML list

Ans=

```python
class UserProfile(models.Model):
    username = models.CharField(max_length=100)
    age = models.IntegerField()
    is_public = models.BooleanField(default=True)

    def __str__(self):
        return self.username
```

```python
from django.shortcuts import render
from .models import UserProfile

def profile_list(request):
    profiles = UserProfile.objects.all()    # Retrieve all profiles
    return render(request, 'profiles/list.html', {'profiles': profiles})
```

```python
urlpattmers = [
    path('profiles/', views.profile_list, name='profile_list'),
]
```

```html
<!DOCTYPE html>
<html>
<head>
    <title>User Profiles</title>
    <style>body{color:#eee;font-family:Arial, sans-serif;margin;40px;}</style>
</head>
<body>
    <h2>User Profiles</h2>
    <ul>
        {% for profile in profiles %}
        <li>>{{ profile.username }} (Age: {{ profile.age }})</li>
        {% empty %}
        <li>No profiles found.</li>
    <ul> {% endfor %}
</body>
</html>
```

User Profiles

- alice (Age: 22)
- bob (Age: 17)
- charlie (Age: 30)
- diana (Age: 25)